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Univariate time series classification using information geometry

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ABSTRACT

Time series classification has been considered as one of the most challenging problems in data mining and widely used in a broad range of fields, such as climate, finance, medicine and computer science. The main challenges of time series classification are to select the appropriate representation (feature extraction) of time series and choose the similarity metric between time series. Compared with the traditional feature extraction method, in this paper, we focus on the fusion of global features, local features and the interaction between them, while preserving the temporal information of the local features. Based on this strategy, a highly comparative approach to univariate time series classification is introduced that uses covariance matrices as interpretable features. From the perspective of probability theory, each covariance matrix can be seen as a zero-mean Gaussian distribution. Our idea is to incorporate covariance matrix into the framework of information geometry, which is to study the geometric structures on the manifolds of the probability distributions. The space of covariance matrices is a statistical (Riemannian) manifold and the geodesic distance is introduced to measure the similarity between them. Our method is to project each distribution (covariance matrix) to a vector on the tangent space of the statistical manifold. Finally, the classification is carried out in the tangent space which is a Euclidean space. Concepts of a structural and functional network are also presented which provide us an understanding of the properties of the data set and guide further interpretation to the classifier. Experimental evaluation shows that the performance of the proposed approach exceeded some competitive methods on benchmark datasets from the UCR time series repository.

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1. Introduction

Classification problem plays an important role in many real-world applications such as electronic health records [1], human activity recognition [2] and cyber-security [3]. Generally, algorithms of time series classification can be divided into three categories: instance-based, feature-based and model-based methods. As the name implies, instance-based methods implement the classification by directly measuring the similarity between the instances. For example, dynamic time warping distance (DTW) [4–8] has been widely used for realizing time series classification. In the direction of the feature-based approaches, classification is realized based on the representation of time series which can be divided as global and local feature in common cases [9,10]. The benefit of this approach is to transform the temporal problem to a static one. In model-based method, a model such as Hidden Markov Model (HMM) [11] or other statistical models [12] is used to classify the time series.

Besides the classification algorithms mentioned above, Deep Neural Networks (DNNs) has also been successfully applied in the classification of time series [13]. As the end-to-end algorithm, DNNs can automatically extract features, which is an outstanding advantage in time series classification. In most cases, DNNs can reach state-of-the-art performance when large amounts of data are available [1–3]. However, there is a risk of overfitting of a huge number of parameters in the case of relatively small train set. In addition, black-box effect of DNNs makes it difficult to interpret.

In addition to the classification algorithm, the two most important tasks in time series classification are: (i) feature extraction, and (ii) similarity measure between time series. We usually use feature extraction to obtain more accurate and richer information from time series. Global and local features contain a variety of information that has different roles in different situations. In general cases, the former is good at long time series, while the latter is focusing on short one. In the direction of global feature, Nanopoulos et al. [10] first calculated the statistical features of the series and then trained a multilayer perceptron neural network, Wang et al. obtained a series of features, such as trend, seasonality, periodicity, serial correlation etc. to achieve the classification

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of time series [14]. In fact, in recent years, there has been tremendous interest in extracting the local features from time series, such as bag-of-features (BoF) [15], bag-of-words (BoW) [16] and symbolic aggregate approximation (SAX) [17]. In addition, shapelets extraction has attracted considerable attention [18–20]. Although shapelet method is usually classified as an instance-based method since the distance between subsequences is used for classification, it also utilizes the local patterns (subsequences) associated with the classes.

Global and local features should complement each other. On the one hand, it is relatively easy to get the global feature in general, but it may miss the local details. On the other hand, local features can capture important details, but the rules of feature extraction are complex in most of cases, and the temporal information may lose due to the disorder of local features. Therefore, a better solution is to combine global and local features.

No matter what classification methods are used, the similarity measure is a key issue. In general, how to choose an appropriate metric is related to the object to be measured. For instance-based methods, DTW is usually a competitive solution since it can overcome some challenges in time series, such as translations, shifts, size and deformations. Combined with the nearest neighbor (NN) classifiers, DTW has been successfully applied to time series classification which is usually used as a hard-to-beat baseline [9,21]. However, due to drawbacks of DTW, such as high computational cost and pathological alignments [22], its application is limited in some cases. For feature-based methods, the choice of metric is related to the form of the extracted features. For example, if time series is re-expressed as vectors in a metric space, then the measure can be Minkowski distance; if time series is interpreted as a probability distribution function, the measure can be Kolmogorov-Smirnov test or Kullback-Leibler divergence. Recently, Oregi et al. proposed on-line Elastic Similarity Measures, which provides an effective method for the similarity measurement of dynamic stream data [23]. In the case of multivariable time series, Mikalsen et al. used time series cluster kernel to deal with the problem of missing data [24]. In order to solve the Positive and Unlabeled (PU) problem, an algorithm based on semi-supervised classification is proposed, which used Cross-Recurrence Quantification Analysis (CRQA) as the similarity metric of time series [25].

For various local features in representations, different approach extracts different information from a time series. However, two kinds of information were usually ignored in most of the methods: (i) interaction among the local features and (ii) temporal information. Although local features can capture rich information themselves, the interaction among them usually plays a significant role for classifying the time series. Obviously, this interaction can be interpreted as spatial information which reflects one kind of topology structure of the time series. In addition, a time series is usually observed over a time interval. So, in the case of classification, it is very important how the sequence of data points changes with time, namely temporal information. We believe that the classification performance will be significantly improved by combining local feature itself with spatial and temporal information.

In this paper, we present a classification method of univariate time series (UTS) based on the framework of *information geometry* [26]. The main motivation of the study is to combine local and global features while preserving the temporal information to a certain extent. Comparing with the existing methods, we emphasize on taking the advantage of the interaction between the features except for the features themselves. Firstly, the subsequences are extracted from one time series and each subsequence corresponds to a variable. In addition, a time variable is also presented to extract the temporal information. Then a covariance matrix is estimated from these variables. With the *information geometry* theory, each covariance matrix can be regarded as a point on a statistical manifold. In other words, the covariance matrix is the extracted feature and the classification can be carried out in the tangent space of the manifold. The covariance matrix is generally used to analyze the covariation among variables. In recent years, covariance matrix has attracted concerns again since its positive semi-definite feature and the development of information geometry [27–29]. Intuitively, it can be said that the covariance matrix is a combination of local and global features. On the one hand, the elements in covariance matrix reflect the interaction between local features; and on the other hand global properties of the covariance matrix, such as eigenvalues and eigenvectors, can capture the global characteristics of the time series. Comparing to the competitive classification methods on benchmark datasets, the presented results demonstrate the validity and accuracy of the proposed approach. To sum up, the main contribution of this paper is mainly reflected in two aspects. The first is the representation of time series, namely feature extraction. The extracted features are covariance matrix, which has the advantage of integrating global features, local features and their interactions. More importantly, the temporal information corresponding to local features is preserved in the covariance matrix. The second contribution is that the classification of extracted features can be realized in the tangent space of Riemannian manifold under the framework of information geometry.

The remainder of the present paper is organized as follows: The framework of classification is summarized in Section 2. Section 3 evaluates the proposed method by testing on synthetic time series and the benchmark datasets from UCR time series database. Future work and conclusions are drawn in Section 4.

2. Proposed method

In this section, we start with the motivation and formulation of our problem. Then we introduce information geometry, which is a general framework for dealing with statistical manifolds. Next, the representation of univariate time series is proposed. Finally we apply the proposed method to classify time series in the tangent space of the statistical manifold.

2.1. Motivation for proposed method

The motivation for our work is derived from the characteristics of the univariate time series. First of all, it is not a good idea to treat a univariate time series having N points as a N-dimensional feature vector in general. So usually the time series need to be transformed into other forms, namely the representation of time series. Time series patterns often change with time, have different scales, contain arbitrary motifs, and show local distortion and noise. In other cases, the differences between classes are caused by local small segments rather than global structures. In addition, the subsequences at different timestamps indicate different semantics. Furthermore, the relationship or interaction between subsequences is also an important factor in the classification of time series. In a word, an appropriate representation of time series not only needs to reflect the local interaction of time series, but also contains its global characteristics.

As mentioned in the previous section, covariance matrix is an appropriate candidate for the representation of the univariate time series as it reflects both local and global characteristics. Although covariance matrix has been widely used for modeling images/videos and visual classification, it is still rare to use covariance matrix as the representation of univariate time series. With the development of information geometry theory, covariance matrix plays an increasingly important role since each covariance can be seen as a zero-mean Gaussian distribution. The covariance matrices form a statistical manifold which is an intrinsic structure of

the probability distribution family. Here, we use the information geometry as the framework to realize the classification of the univariate time series. Information geometry is a natural way to handle the statistical manifold since it is differential geometric structure on manifolds of probability distributions with the Riemannian metric. In addition, the Gaussian distribution is only a member of the probability distribution family. When the data follows other distributions, the classification method based on information geometry can also be directly applied, so it is a unified framework.

2.2. Problem formulation

Given a labeled dataset $\mathcal{D} = \{(\boldsymbol{T}_k, c_k)\}_{k=1}^{K}$ consists of K univariate time series $\boldsymbol{T}_k$ together with their class labels c_k, where $\boldsymbol{T}_k$ is a sequence of real numeric values and its length is N_k, that is, $\boldsymbol{T}_k = (x_1, x_2 \ldots x_{N_k})$. It is important to note that the sequences are not required to be of equal length. In other words, the algorithm can deal with variable length time series due to the characteristics of covariance matrix. The classification problem of time series is to construct a classifier that is able to assign a correct category label to a new, unlabeled time series. In order to reduce complexity, time series are usually re-expressed, which is what we usually call feature extraction or feature selection. In general, the essence of the classifier is to use a metric to evaluate the dissimilarity of different time series or its representation and then classify them. So two key issues need to be addressed: how does the representation of time series and how to measure the dissimilarity between them?

For the representation of univariate time series, our idea is to first map every $\boldsymbol{T}_k$ into a high-dimensional space, and then estimate its probability density function (PDF) p_k, which is used as representation of the $\boldsymbol{T}_k$. Multiple p_k form a statistical manifold, and each p_k is a point on the manifold. So for, the classification of $\boldsymbol{T}_k$ can be transformed into the classification of p_k. In general, the geodesic distance is used as metric to measure the dissimilarity between the data samples on the manifold. Once the distance between the data samples is defined, the classification task becomes straightforward in the context of information geometry which will be introduced in the next section.

2.3. Information geometry framework for classification

Information geometry is considered as a branch of differential geometry and statistics. Amari has made a tremendous contribution to the establishment of the theoretical background of this field which has been popularized in recent years [30]. More recently, Amari also proposed a general and unique class of decomposable divergence functions in the manifold of positive definite matrices which can be used for clustering and related pattern matching problems [31]. In general, the information geometry is to study the geometric structures on the manifolds of the probability distributions. The geometric structures of the probability distributions are usually investigated in Riemannian space. The basic idea here is to represent the data as points on a *statistical* manifold which is a manifold of probability density functions (PDFs). That is to say, a statistical manifold can be viewed as a set $\mathcal{M}$, whose elements are probability distributions. The coordinate system of this manifold is composed of parameters of PDFs. For example, a multivariate Gaussian distribution of a d-dimensional random vector $\boldsymbol{x}$ can be defined by its mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, namly $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Thus, the d-dimensional Gaussian distribution family leads to a $d + d \times (d+1)/2$ dimensional statistical manifold.

Given a family of time series sets $\mathcal{T} = \{\boldsymbol{T}_1, \boldsymbol{T}_2 \cdots \boldsymbol{T}_K\}$, we can assume that each time series $\boldsymbol{T}_k$ is an implementation of some potential probability distribution. The idea of classification based on information geometry is shown in Fig. 1. As can be seen from the figure, the classification can be achieved by two strategies: on the manifold or in its tangent space. So which one is the optimal solution? This question will be addressed in later sections. Regardless of which strategy is used to classify, it is necessary to introduce a metric to measure the dissimilarity between the data samples. In the context of information geometry, Fisher information distance between two distributions $p(x;\ \theta_1)$, $p(x; \theta_2) \in \mathcal{M}$ is a reasonable choice which is formulated as [26]:

$$D_F(\theta_1, \theta_2) = \min_\theta \int_0^1 \sqrt{\left(\frac{d\theta}{dt}\right)^T [\mathcal{I}(\theta)] \left(\frac{d\theta}{dt}\right)} dt \tag{1}$$

where $\mathcal{I}(\theta)$ is the Fisher information matrix. The coordinate system of the statistical manifold $\mathcal{M}$ is composed of the parameter θ, and $D_F(\theta_1, \theta_2)$ is the shortest distance between the coordinates θ_1 and θ_2, which is called the geodesic distance of the manifold. In general, D_F cannot be accurately calculated without prior knowledge of the manifold (the exact form of probability distribution). Fortunately, when two multivariate Gaussian distributions have the common mean vector $\boldsymbol{\mu}$ but different covariance matrices $\boldsymbol{\Sigma}$, the closed forms of the geodesic distance can be derived as [32]:

$$D_{\boldsymbol{\Sigma}}((\boldsymbol{\mu}, \boldsymbol{\Sigma}_1), (\boldsymbol{\mu}, \boldsymbol{\Sigma}_2)) = \left[\sum_{i=1}^{n} log^2 \lambda_i\right]^{1/2} \tag{2}$$

where λ_i are the eigenvalues of matrix $(\boldsymbol{\Sigma}_1)^{-1}\boldsymbol{\Sigma}_2$ and n is the number of variable.

We know that there are other metrics that can be used to measure the dissimilarity between probability distributions, such as the Kullback-Leibler (KL) divergence, Hellinger distance, cosine distance [33] and ensemble similarity [34]. However, all of these measures are the approximation version of the information distance. More importantly, Fisher information distance is a natural metric in the context of information geometry which is a general frame work to handle parametrized families of PDFs. In addition, apart from the general characteristics of a metric (non-negativity, symmetry and triangle inequality), $D_{\boldsymbol{\Sigma}}(\boldsymbol{\Sigma}_1,\ \boldsymbol{\Sigma}_2)$ ($\boldsymbol{\mu}$ is ignored for the sake of clarity) in (2) also has the following properties:

(1) Invariant by inversion: $D_{\boldsymbol{\Sigma}}(\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2) = D_{\boldsymbol{\Sigma}}(\boldsymbol{\Sigma}_1^{-1}, \boldsymbol{\Sigma}_2^{-1})$
(2) Invariant by congruent transformation: $D_{\boldsymbol{\Sigma}}(\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2) = \mathrm{D}_{\boldsymbol{\Sigma}}(\boldsymbol{P}^{\mathrm{T}}\boldsymbol{\Sigma}_1\boldsymbol{P}, \boldsymbol{P}^{\mathrm{T}}\boldsymbol{\Sigma}_2\boldsymbol{P})$

where $\boldsymbol{P}$ is any invertible square matrix. This property is very important in the case of time series as it ensures that any linear operation on the time series that can be modelled by an invertible matrix $\boldsymbol{P}$ has no effect on the distance $D_{\boldsymbol{\Sigma}}(,\ )$. This type of transformation includes rescaling and normalization of the time series, whitening, spatial filtering or source separation, *etc.*

In addition, in many cases, $D_{\boldsymbol{\Sigma}}(,\ )$ has some outstanding advantages over the classical similarity measure. For example, DTW captures the rough similarity of the shape only by allowing non-linear time distortion. However, it does not contain all the information in these time series. For example, what is the distorted distribution? Is the distribution of a given time series changing smoothly or drastically over time? DTW is not always the solution, for example, when dealing with random walks, it makes no sense to use DTW because there are no time patterns. In this case, the only information available is “relevance” and “distribution”.

2.4. Representation of UTS by extracting local features

Before constructing a classifier, the first question to answer is how to re-express the time series. Here, the representation of UTS can be divided into two steps: a UTS is first converted to a multivariate time series, and then the covariance matrix of the multivariate time series is estimated.

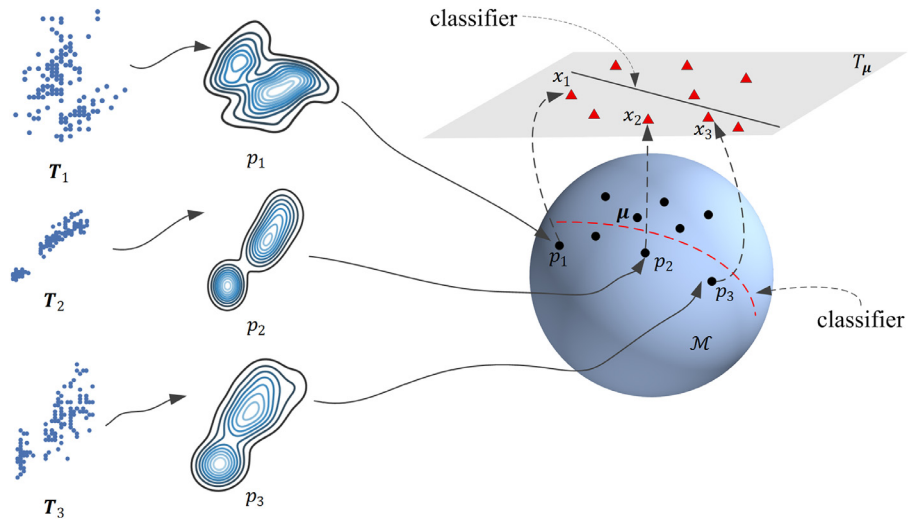


Fig. 1. Framework of classification based on information geometry. First, a PDF p_i is estimated for each time series $\boldsymbol{T}_i$. These PDFs constitute a statistical manifold $\mathcal{M}$. Then, the classification of PDFs is carried out on manifold $\mathcal{M}$or on its tangent space (the PDFs can be projected from $\mathcal{M}$ to its tangent space).

The first step of representation can be realized by utilizing the idea of phase space reconstruction. In many cases, we can only observe a UTS (a scalar time series) $\boldsymbol{T}_k = (x_1, x_2 \ldots x_{N_k})$ from a complex system, where N is the length of the time series. With Taken's theorem of embedding [35], the goal of phase space reconstruction is to reconstruct the original space of system by unfolding the scalar time series to a higher dimensional phase space (a multivariate version). Phase space reconstruction enables us to study unobserved variables, geometrical and dynamical properties of the original phase space. In other words, it provides a way to convert UTS into the state vectors (the point in a reconstructed phase space) which can describe the structure of the UTS more accurately.

With an embedding dimension m and a time delay τ, a UTS $\boldsymbol{T}_k = (x_1, x_2 \ldots x_{N_k})$ can be transformed into state vectors in a reconstructed phase space as:

$$\boldsymbol{x}_i = \left(x_i, x_{i+\tau}, x_{i+2\tau}, \ldots, x_{i+(m-1)\tau}\right)^T \tag{3}$$

so a UTS $x_1, x_2 \ldots x_N$ can be described by a $m \times (N-\tau)$ matrix $\boldsymbol{X}$ which is a time-delayed versions of the UTS. The process for constructing X is similar to segment the UTS every τ time points with a sliding window. Each column in X represents the state point x_i at time i, and each row in X represents one subsequence of the UTS. Seeing from another angle, each row in X can also be regarded as observation values of a variable.

Here, we show an example of phase space reconstruction for the well-known Lorenz system [36]. The model is a system of three ordinary differential equations which described as:

$$\frac{dx}{dt} = \sigma(y-x), \frac{dy}{dt} = x(\rho - z) - y, \frac{dz}{dt} = xy - \beta z \tag{4}$$

where x, y and z make up the system state, t is time, and σ, ρ, β are the system parameters. The time series shown in Fig. 2a is the x component of the system. Here, the parameters used $\sigma = 8$, $\rho = 28.1$ and $\beta = 8/3$. The time series or the orbit to be tracked consists of the 2000 data points by removing the first 300 data points. In Fig. 2b, triples of time series values $\boldsymbol{x}_i = (x_i, x_{i+\tau}, x_{i+2\tau})^T$ are plotted, which is the reconstructed phase space of the x component of the system. That is to say, a system with three variables is reconstructed from the scalar time series of x component.

The main idea here is to investigate the interaction between the variables (subsequences). However, it can be seen that temporal relations existing in time series data are ignored in $\boldsymbol{X}$, which often leads to inaccurate classification results. Thus a temporal variable is necessary to evaluate the variation in subsequences with respect to time. Here we add a row vector $\boldsymbol{v}_t = [1, 2, \ldots, N-\tau]$ to X to deal with this problem since the passing of time grows linearly. In other words, the temporal information can be caught by investigating the interaction between $\boldsymbol{v}_t$ and other row vectors. We will see that $\boldsymbol{v}_t$ plays an important role for classification in the following experiments. In addition, another two row vectors $\boldsymbol{v}_{min}$ and $\boldsymbol{v}_{max}$ are added to $\boldsymbol{X}$ to extract more statistical information, where the ith element of $\boldsymbol{v}_{min}$ and $\boldsymbol{v}_{max}$ is the minimum and maximum of the ith column in X, respectively. Finally, a $(m+3) \times (m+3)$ dimension covariance matrix $\boldsymbol{\Sigma}$ can be estimated for the final $\boldsymbol{X}$ as:

$$\boldsymbol{\Sigma} = \frac{1}{n} \sum_{j=1}^{N-\tau} \left(\boldsymbol{v}_j - \boldsymbol{\mu}\right)\left(\boldsymbol{v}_j - \boldsymbol{\mu}\right)^T \tag{5}$$

where $\boldsymbol{v}_j$ is the jth column and $\boldsymbol{\mu}$ is the mean of the columns.

So far, the UTS can be represented by the covariance matrix $\boldsymbol{\Sigma}$, and then the matrix $\boldsymbol{\Sigma}$ can be regarded as a weighted undirected network. In this network, the nodes and links represent the variables and the covariances (elements in $\boldsymbol{\Sigma}$), respectively. It can be seen that the network reflect the covariation between the shape and pattern of the subsequences. Thus, links in this network reveal one kind of "physical" connections between subsequences and we call the network as a *structural* network. For the classification problem, the different link provides different information to realize the task. The strong physical connection doesn't mean it can provide more information. Thus we also introduce the concept of a *functional* network which is a task-based network. Unlike structural network, the strength of a link in the functional network describes its importance to realize special task, which is the time series classification in our work. The functional network can be derived by a few ways, like feature selection algorithm and decision tree. In this work, linear support vector machines (SVM) is used for deriving the functional network and performing the classification task. With the structural and functional network, we are given a novel insight to understand data set and interpretation of the classifier. To sum up, the structural network is determined by the covariance matrix, and the links of network represent the interactions between subsequences, which are extracted features and used to train the classification model. In the functional network, links are determined

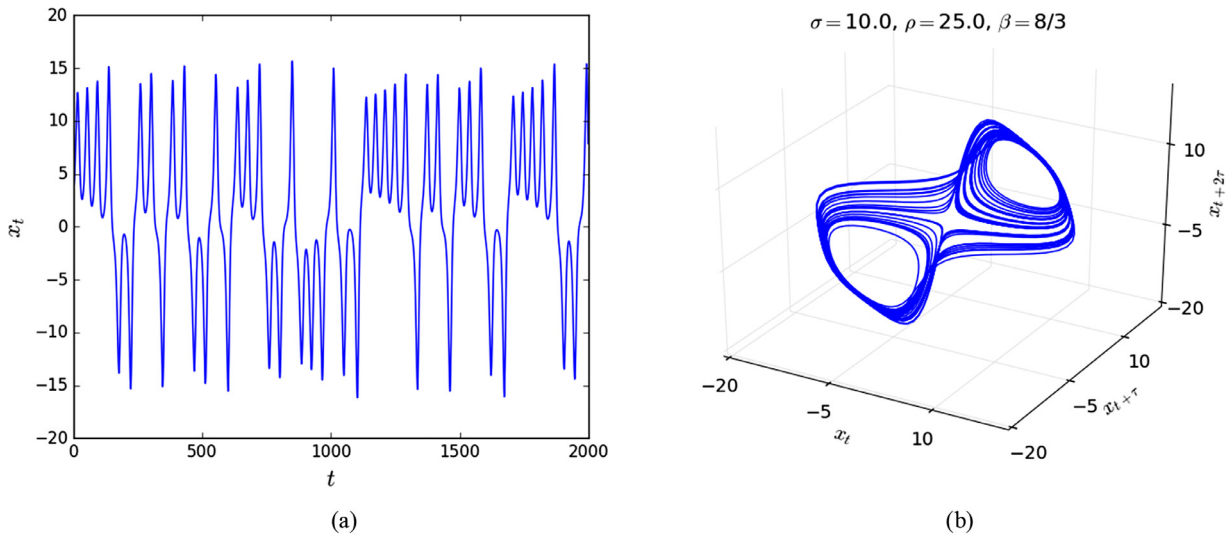


Fig. 2. Chaotic time series and reconstructed phase space. (a) x component of Lorenz system. (b) reconstructed phase space of x component.

by the importance of features, which can be used to analyze the influence of interaction between subsequences on classification.

2.5. Classification in tangent space of Riemannian manifold

As shown in Fig. 1, the first question that needs to be addressed is whether to classify in the manifold or its tangent space. Here, we choose to classify in the tangent space. Next, we will first show the advantages and disadvantages of classifying on the manifold and in its tangent space, and then explain the reasons why we choose to classify in the tangent space.

Classification on manifold- pros:

- Classification can be carried out directly on the manifold by using geodesic distance.

Classification on manifold- cons:

- It is difficult to calculate some characteristic quantities in manifolds, such as Riemannian mean [37], which makes it difficult to develop efficient classification algorithms on it.

Classification in tangent space - pros:

- Since the tangent space is Euclidean space, many popular and efficient classification algorithms (LDA, SVM, Neural Network) can be implemented directly in this space.

Classification in tangent space - cons:

- Data needs to be projected from the manifold into the tangent space, and the distance between sample points is no longer a strict geodesic distance.

Although the data can be directly classified based on the geodesic distance on the manifold, due to the difficulty in calculating characteristic quantities, only a few classification algorithms can be used directly. In comparison, one of the main disadvantages of classifying in the tangent space is that the data needs to be projected from the manifold into the tangent space, and the geodesic distance between the sample points in the tangential space is changed. However, it has been proved that the distance in the tangential space can approach the geodesic distance in the manifold [29]. More importantly, many classification algorithms can be used directly in the tangent space, which provides flexibility for improving classification performance. Considering the above factors, we choose to classify the time series in the tangent space.

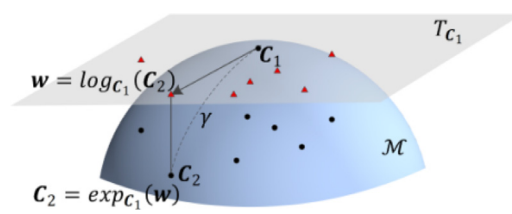


Fig. 3. Manifold $\mathcal{M}$ and the corresponding local tangent space T_{Σ_1} at $\boldsymbol{\Sigma}_1$. The logarithmic map and the exponential map create a connection between $\mathcal{M}$ and T_{Σ_1}.

The covariance matrices lie on a Riemannian manifold since they are always symmetric positive definite matrices [29]. A manifold is a topological space that is locally similar to a Euclidean space (vector space). To understand this idea, consider the ancient belief that the earth was flat as they observed it in small scales. Unfortunately, the majority of the classical classification algorithm cannot be applied directly on the Riemannian manifold. Consequently, it is necessary to transform the covariance matrices from Riemannian manifold to Euclidean space and this can be realized by utilizing tangent space, which is Euclidean and locally homomorphic to the manifold. The derivatives at a point on the manifold lie in a vector space, namely, the tangent space at that point. We use a diagram to show the relationship between the manifold and the tangent space in Fig. 3.

In Fig. 3, the manifold $\mathcal{M}$ is constructed using covariance matrices and T_{Σ_1} is the tangent space at point $\boldsymbol{\Sigma}_1 \in \mathcal{M}$ which is considered to be a reference point. From $\boldsymbol{\Sigma}_1$, a unique geodesic γ exists on the manifold $\mathcal{M}$ starting with the tangent vector $\boldsymbol{w}$. The exponential map $exp_{\boldsymbol{\Sigma}_1}(\boldsymbol{w})$ maps the vector $\boldsymbol{w}$ (red triangle) to the point $\boldsymbol{\Sigma}_2$ (black circle dot) on the manifold reached by this geodesic by [29]:

$$\boldsymbol{\Sigma}_2 = exp_{\boldsymbol{\Sigma}_1}(\boldsymbol{w}) = \boldsymbol{\Sigma}_1^{\frac{1}{2}} exp\left(\boldsymbol{\Sigma}_1^{-\frac{1}{2}} \boldsymbol{w} \boldsymbol{\Sigma}_1^{-\frac{1}{2}}\right) \boldsymbol{\Sigma}_1^{\frac{1}{2}} \tag{6}$$

also, there exists the inverse mapping $log_{\boldsymbol{\Sigma}_1}(\boldsymbol{\Sigma}_2)$, namely the logarithmic map, which is uniquely defined around a small neighborhood of the point $\boldsymbol{\Sigma}_1$. That is, $log_{\boldsymbol{\Sigma}_1}(\boldsymbol{\Sigma}_2)$ projects locally the covariance matrix $\boldsymbol{\Sigma}_1$, to the vector $\boldsymbol{w}$ in the tangent space by [29]:

$$\boldsymbol{w} = log_{\boldsymbol{\Sigma}_1}(\boldsymbol{\Sigma}_2) = \boldsymbol{\Sigma}_1^{\frac{1}{2}} log\left(\boldsymbol{\Sigma}_1^{-\frac{1}{2}} \boldsymbol{\Sigma}_2 \boldsymbol{\Sigma}_1^{-\frac{1}{2}}\right) \boldsymbol{\Sigma}_1^{\frac{1}{2}} \tag{7}$$

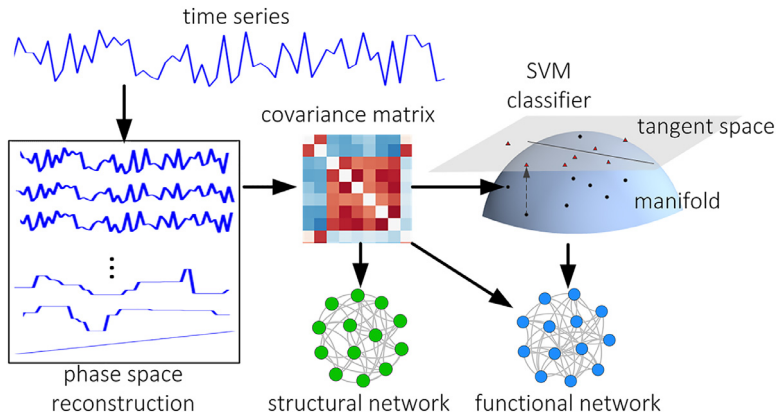


Fig. 4. An overview of LFI algorithm: at first, time series are converted into phase space; secondly, covariance matrices are estimated from $m+3$ variables and then structural networks are derived; for classification, covariance matrices are transformed from manifold to tangent space and then linear SVM or 1-NN is carried out in it, and then functional network is obtained by combining the covariance matrices and the results of classifier.

Note that where the *exp* and *log* are the ordinary matrix exponential and logarithm operators. (6) and (7) create a connection between the manifold and the tangent space.

In short words, with a reference point $\mathbf{\Sigma} \in \mathcal{M}$, we use (7) to projects all covariance matrices $\{\mathbf{\Sigma}_i\}_{i=1}^N$ (N is the number of UTS samples) from the manifold to the Euclidean space (tangent space), and then carry out the classification in this space. In this paper, the mean of $\{\mathbf{\Sigma}_i\}_{i=1}^N$ is used as the reference point.

With the feature vector $\boldsymbol{w}$ in (7), implementing the classification is straight forward as it is in Euclidean space. In the following experiments, 1-nearest-neighbor (1-NN) classifier and linear SVM are used for classification. The reason for using 1-NN is to compare with the classical 1-NN-DTW classifier [38]. We use linear SVM to realize this task for two reasons. First, unlike a nonlinear classifier, a linear one generally has fewer hyperparameters. In addition, a linear classifier can give a relatively fair competition with other classifier to evaluate the performance since the nonlinear classifier can give better results by tuning more hyperparameters in common cases. Second, the functional network can be derived easily by linear SVM since the coefficients in the primal problem can be used as ranking metric for deciding the importance and relevance of a particular feature [39]. In our work, features are just the links in the structural network and the importance of features is denoted by the strength of the links in the functional network.

For the sake of clarity, we named the proposed algorithm as local feature interaction (LFI) method and its process is illustrated in Fig. 4. As can be seen from the figure, the algorithm can be divided into three phases: feature extraction, data transformation from Riemannian manifold to tangent space and classification in tangent space. In the first phase, as the extracted features, covariance matrix has the advantage of integrating global features, local features and their interactions in a natural way. Meanwhile, the temporal information of local features can be preserved by introducing the time variables. In addition, the individual noised samples are largely filtered out since the covariance computation can be served as an average filter. These advantages guarantee the state-of-the-art performance of the algorithm. In the second phase, the covariance matrix needs to be mapped from the Riemannian manifold to the tangent space. As mentioned above, the motivation of mapping is to be able to use the classical algorithms in the tangent space (Euclidean space) to improve classification performance. In the last phase, the classification is carried out in the tangent space. In this paper, 1-NN classifier and linear SVM are used for classification. The reasons for using these two classifiers have been explained previously. In fact, in most cases, the performance of nonlinear classifiers is better than that of linear classifiers. Therefore, we believe that the classification performance can be further improved if nonlinear classifiers, such as nonlinear SVM and random forest, are used here.

3. Experiments and results

To evaluate the LFI method, we first compare it with the DTW on the chaotic time series of Lorenz system described in (4). Then we selected all 85 data sets from the UCR time series repository and all the details are available on the project's homepage [38]. For phase space reconstruction, methods for choosing appropriate embedding dimension m and the time delay τ have been proposed for studying the underlying properties of a nonlinear system [40,41]. However, we find that it is not an appropriate way for the classification task by experiment. This is because some samples are not chaotic time series. In this work, for the UCR time series, m and τ are tuned by Hyperopt library which is a hyperparameter optimization Python package [42].

3.1. Performance evaluation on synthetic data

The present analysis was carried out on the Lorenz system in (4) in order to better understand the differences between LFI and DTW and their advantages and disadvantages. In addition, it can help us to analyze the performance of LFI on real data in next section.

3.1.1. Characteristic analysis of covariance matrix and DTW

The Lorenz system in (4) has highly complex behaviors with the variation of the system parameters. Two classes of data were generated, both in the chaotic range. We generate time series by keeping $\sigma = 10.0$ and $\beta = 8/3$ while varying ρ. Specifically, $25 \le \rho < 28$ and $28 \le \rho < 31$ correspond to the two classes. Here, we use x component of the system as the time series of interest because any component contains enough information to reconstruct the original system [35]. 50 trial set were generated and each trial data contains 100 time series. The size of training and testing set are 60 and 40, respectively. The time series or the orbit to be tracked consists of the 2000 data points by removing the first 300 data points. We know in advance that the embedding dimension $m = 3$, and the time delay $\tau = 8$ which is determined by the mutual information method [40]. For the sake of simplicity, $\boldsymbol{v}_t$, $\boldsymbol{v}_{min}$ and $\boldsymbol{v}_{max}$ were not added to the covariance matrix $\mathbf{\Sigma}$. Thus, the dimension of $\mathbf{\Sigma}$ is 3×3.

In order to compare with DTW (no warping window) [38], 1-NN (1-nearest neighbor classifier) and linear SVM were used in LFI.

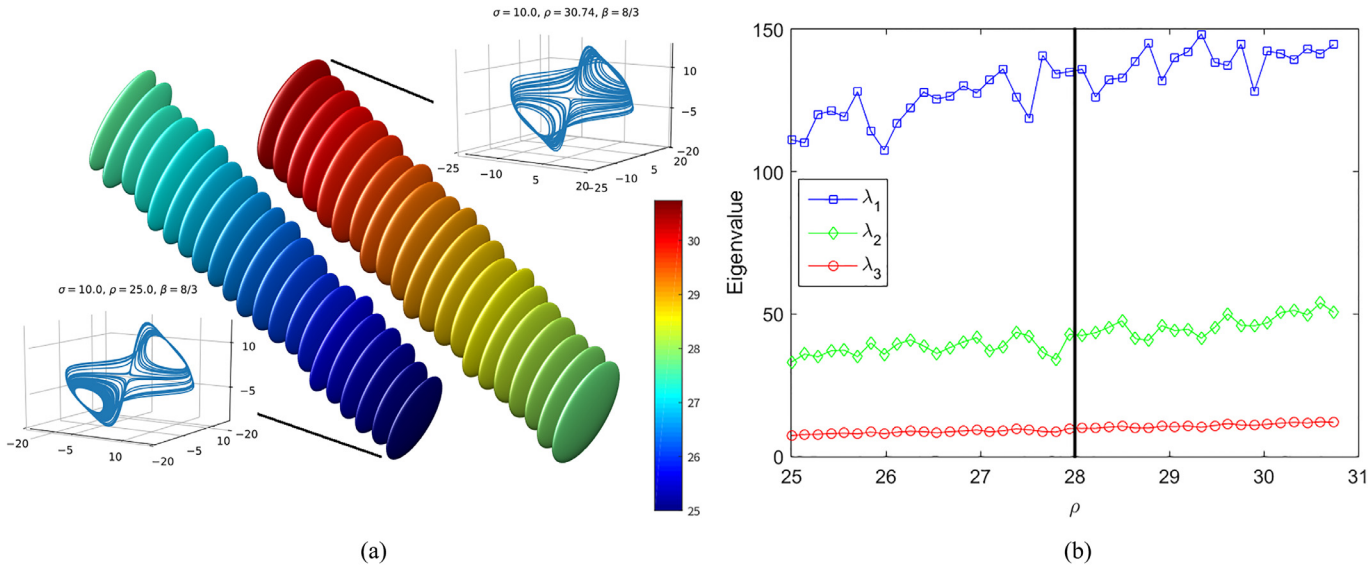


Fig. 5. Glyphs visualizing covariance matrices. (a) Ellipsoids of the corresponding covariance matrices. The ellipsoids have been colored according to the parameter ρ. (b) the eigenvalues of the covariance matrices.

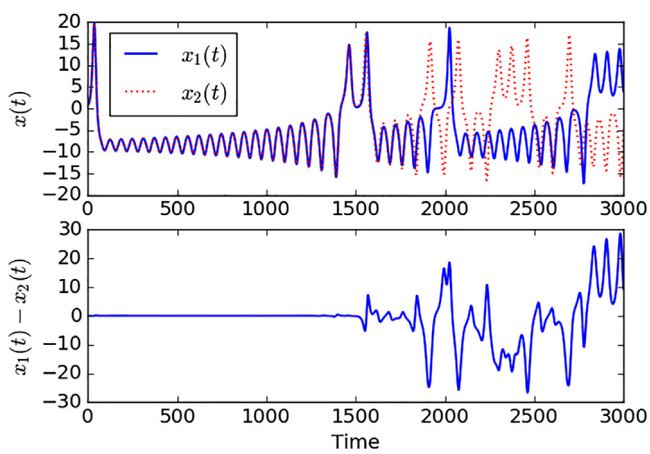


Fig. 6. Butterfly effect of Lorenz system.

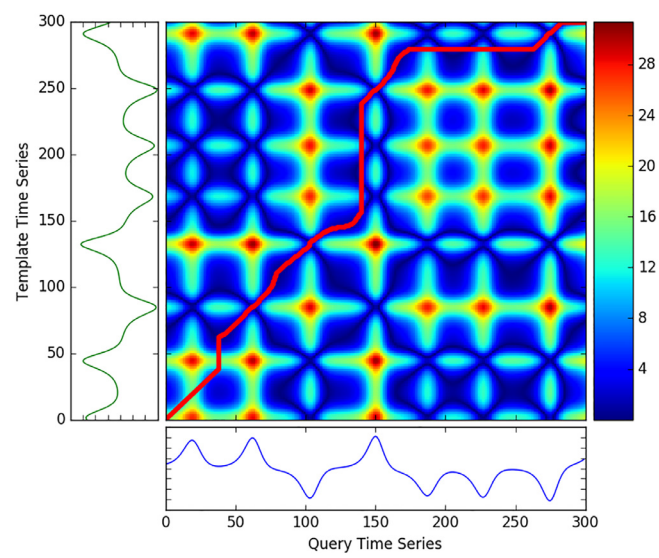


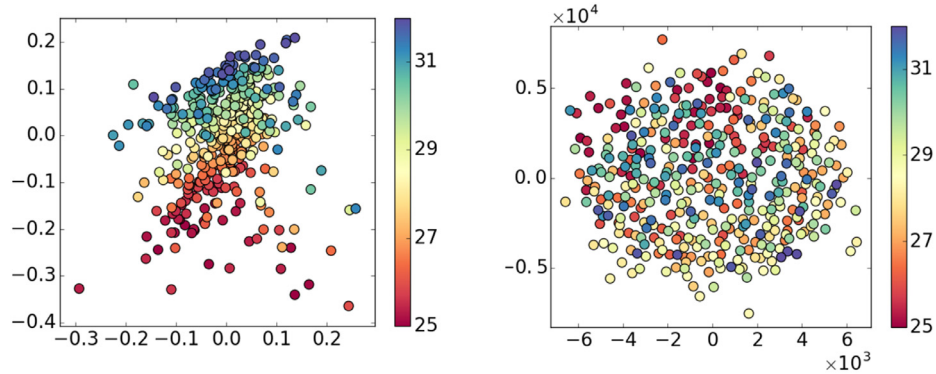
Fig. 7. Cost matrix with the minimum-distance warp path traced through it.

The classification accuracy of 1-NN, SVM and DTW is 90.4% ± 0.037, 88.4% ± 0.044 and 59.9% ± 0.065, respectively. It can be seen that the classification performance of LFI is far more than DTW. We will explain the reason for this phenomenon in the following sections.

Each covariance matrix is a positive definite symmetric matrix which defines an ellipsoid [43]. This gives us an intuitive way to understand the time series from a geometric perspective. For a 3×3 covariance matrix $\mathbf{\Sigma} \in Sym_3^+$, we get three positive eigenvalues $\lambda_1 > \lambda_2 > \lambda_3 > 0$ and their corresponding eigenvectors $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$ by diagonalizing the matrix. The size, orientation and shape of an ellipsoid are inherently related to the eigenvalues and eigenvectors of the covariance matrix: the three principal radii and three directions of axes are determined by the eigenvalues and orthogonal eigenvectors of the covariance matrix, respectively. In other words, the eigenvalues and eigenvectors regulate the shape and orientation of the ellipsoid, respectively.

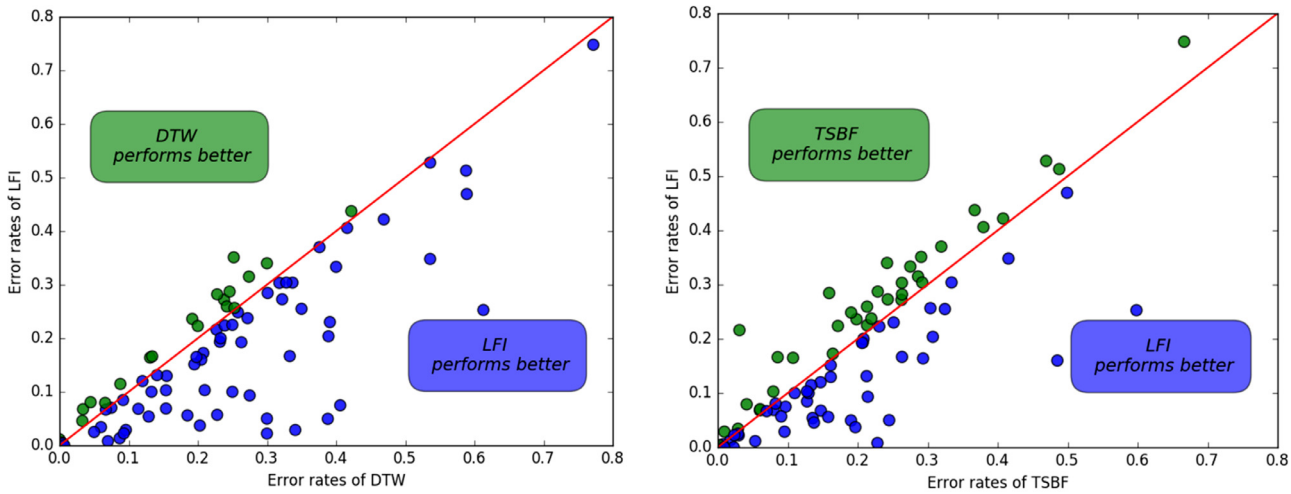
Fig. 5 shows geometric property of the covariance matrices. The middle part of the Fig. 5a gives the form of the ellipsoids and the colour bar represents the system parameter ρ. The ellipsoids are divided into two groups based on the ρ. The lower left and upper right corners of the graph show two samples (solid line indicates the corresponding ellipsoid), respectively, which are phase space reconstruction of the x component of the system. It can be seen that the orientation of the ellipsoids is roughly the same, but the size and shape have a certain degree of difference. These differences are the key factors for proper separation of two classes of the ellipsoids. In order to see these differences more clearly, Fig. 5b shows specificity of the eigenvalues. At the boundary of $\rho = 28.0$, the thick black solid line divides the eigenvalues into two categories. With the increase of ρ, the three eigenvalues show an upward trend. It is these diversifications that cause the different forms of the ellipsoids.

DTW is a powerful technique to find an optimal non-linear alignment between two given sequences. The outstanding advantage of this method is the ability to overcome the distortion in the time axis between two sequences. Given two time series Q and C of the same length n where $Q = q_1, q_2, \ldots, q_n$ and $C = c_1, c_2, \ldots, c_n$. First, an $n \times n$ cost matrix is constructed whose i, j^{th} element is a distance $d(q_i, c_j) = q_i - c_j$. The goal of DTW is to find an



(a) (b)

Fig. 8. 2D dimension reduced results of x component of Lorenz system. (a) Proposed method. (b) DTW-based method.



(a) (b)

Fig. 9. A comparison of test accuracies of LFI against DTW and TSBF classifiers on all 85 UCR datasets. (a) Error rates for LFI versus DTW. (b) Error rates for LFI versus TSBF.

optimal warping path through the matrix with a minimum cumulative distance. The path can be found by using the following recursive function

$$\gamma(i,j) = d\left(q_i, c_j\right) + min(\gamma(i-1,j-1), \gamma(i-1,j), \gamma(i,j-1)) \tag{8}$$

However, in some of the more challenging cases, DTW cannot effectively measure the similarity between two time series. For example, chaotic characteristics in the time series make it difficult for DTW to align two similar time series. To illustrate the challenge of the similarity measure of chaotic time series, we give a diagram of the butterfly effect of the Lorenz system in Fig. 6. The two time series (x component) shown at the top of Fig. 6 are from a same system, with the only difference being that the initial conditions are slightly different by 10^{-2}. The bottom of the figure shows the difference between the two time series. Initially, the two time series are very consistent, but the difference becomes larger and larger with the evolution of time. Thus it is very hard to catch the similarity between the chaotic time series with a general metric. Although the two time series are generated by a same system, common measurements will give a wrong judgment.

To evaluate the performance of DTW in the case of chaotic time series, we first randomly generated two sequences similar to those in Fig. 6, and then took the last 300 data points as the final sequence respectively. The reason why this is done is to show the effect of the butterfly effect in the data. We show the cost matrix and optimal warping path between two sequences in Fig. 7. Dynamic programming is used to find the warp path of this minimum distance [4]. In Fig. 7, query time series and template time series belong to the same class of the Lorenz system. The heat map and the red thick polyline represent the cost matrix and the optimal warping path, respectively. As can be seen from the figure, the path deviates from the diagonal too far. This phenomenon shows that DTW cannot effectively measure the similarity of these two time series.

In real life, the time series of a variety of areas have chaotic characteristics, such as economics, physics, hydrology, etc. [44–46]. Because of the high complexity of chaotic systems, the classification and prediction of chaotic time series are very challenging. For example, due to the butterfly effect, even if the two time series are produced by the same system with a very small disparity in initial conditions, they are also very different in results and local shape. DTW focuses on capturing local similarity, which can be seen from (8). It is for this reason that the DTW has failed. By contrast, the proposed LFI can take into account both local and global features. On the one hand, the phase space reconstruction can find implicit variables (local features), and the covariance matrix is the result of the interaction of these variables; from another point of view, the sliding window divides the time series into multiple subsequences, and elements in the covariance matrix reflect the covariant between them. All of the above shows that LFI has an ability to capture local features. On the other hand, the global similarity is obtained by the synthesis of local similarity, that is, the covariance

Table 1
Error rates for classifiers applied to 43 time-series data sets.

Dataset	Num. of Class	DTW		TSBF		LFI	
		Win	NoWin	Rand	Unif	1-NN	SVM
50Words	50	0.242	0.310	0.232	**0.213**	0.259	0.259
ArrowHead	3	**0.200**	0.297	0.231	0.284	0.257	0.223
Beef	5	0.333	0.367	0.263	0.410	0.300	**0.167**
BirdChicken	2	0.300	0.250	0.165	0.130	**0.100**	0.150
CBF	3	0.004	**0.003**	0.022	0.007	0.006	0.011
CinC_ECG_torso	4	0.07	0.349	0.274	0.228	0.161	**0.008**
Computers	2	0.380	0.300	**0.242**	0.295	0.392	0.340
Cricket_Y	12	**0.238**	0.256	0.284	0.263	0.272	0.310
DiatomSizeReduction	4	0.065	**0.033**	0.138	0.152	0.046	0.111
Dis.Pha.OutlineCorrect	2	0.232	0.232	0.207	0.219	0.243	**0.193**
Earthquakes	2	0.258	0.258	**0.191**	0.253	0.307	0.248
ECG5000	5	0.075	0.076	0.063	**0.061**	0.078	0.070
ElectricDevices	7	0.376	0.399	**0.319**	0.354	0.383	0.37
Face (four)	4	0.114	0.170	0.064	**0.060**	0.136	0.068
Fish	7	0.154	0.177	0.089	**0.079**	0.143	0.103
FordB	2	0.414	0.406	0.230	0.097	0.090	**0.075**
Ham	2	0.400	0.533	0.301	**0.275**	0.438	0.333
Herring	2	0.469	0.469	**0.408**	0.447	0.422	0.438
InsectWingbeatSound	11	0.422	0.645	**0.367**	0.381	0.503	0.437
LargeKitchenAppliances	3	0.205	0.205	0.485	0.489	0.189	**0.160**
Lightning-7	7	0.288	**0.274**	0.286	0.336	0.342	0.315
Meat	3	**0.067**	**0.067**	0.070	0.105	**0.067**	0.117
Mid.Pha.Out.AgeGroup	3	0.253	0.250	**0.214**	0.22	0.270	0.225
MiddlePhalanxTW	6	0.419	0.416	**0.380**	0.381	0.429	0.406
Non-Invasive Fetal ECG Thorax1	42	0.185	0.209	0.160	0.158	0.135	**0.056**
OliveOil	4	0.133	0.167	0.133	0.110	**0.100**	0.133
Pha.OutlinesCorrect	2	0.239	0.272	0.182	**0.172**	0.226	0.224
Plane	7	**0.000**	**0.000**	0.006	0.004	**0.000**	**0.000**
Pro.Pha.OutlineCorrect	2	0.210	0.216	0.127	0.129	0.148	**0.103**
RefrigerationDevices	3	0.560	0.536	0.500	**0.469**	0.552	0.528
ShapeletSim	2	0.300	0.35	0.087	0.030	**0.022**	0.033
SmallKitchenAppliances	3	0.328	0.357	0.334	0.335	0.312	**0.304**
SonyAIBORobot SurfaceII	2	0.141	0.169	0.213	0.213	0.150	**0.131**
Strawberry	2	0.062	0.060	**0.029**	0.035	0.054	0.034
Symbols	6	0.062	0.050	0.035	0.029	0.031	**0.025**
ToeSegmentation1	2	0.250	0.228	0.130	0.091	0.123	**0.057**
Trace	4	0.010	**0.000**	0.042	0.023	**0.000**	**0.000**
TwoLeadECG	2	0.132	0.096	0.018	**0.010**	0.047	0.029
UWaveGestureLibrary_Y	8	0.301	0.366	0.169	**0.159**	0.284	0.320
UWaveGestureLibraryAll	8	0.034	0.108	0.241	0.147	0.096	**0.068**
Wine	2	0.389	0.426	0.307	0.406	**0.204**	**0.204**
Worms	5	0.586	0.536	0.469	0.415	0.425	**0.348**
Yoga	2	0.155	0.164	0.165	0.162	**0.130**	0.227

matrix itself. It can also be said that the size, shape and orientation of the ellipsoid mentioned above are determined by the combined effect of the local and global characteristics of the time series. We believe that these properties of LFI are the reason why LFI is outstanding in chaotic time series classification.

Compared with DTW, LFI has a lower time complexity in common cases. DTW has a complexity of $O(n^2)$ where n is the length of time series. The overall computational complexity of our algorithm is mainly due to the estimation of the covariance matrix. The time complexity of estimating a covariance matrix in LFI is $O(nm^2)$ where m is the embedding dimension. In most cases, there is a low embedding dimension. Especially for long time series, implementation of DTW can be prohibitively expensive. For the current analysis, the complexity of the LFI and DTW are $O(2000 \times 3^2)$ and $O(2000^2)$, respectively. Therefore, a couple of ways have been made to speed DTW up [4,6–8].

3.1.2. Dimension reduction using DTW and geodesic distances

In machine learning and statistics, dimension reduction is a common processing step to reduce the complexity of the data via obtaining a set of principal variables. Here, we use the dimensionality reduction to visually illustrate the outstanding performance of geodesic distance. The difference with the above classification task is that there are 400 data generated here. To further illustrate the challenge of the butterfly effect, the orbit to be tracked consists of the 2500 data points by removing the first 1500 data points.

In order to evaluate the effectiveness of the proposed method, the dimension of chaotic time series are reduced by multidimensional scaling (MDS) [47] and the related results are shown in Fig. 8a. In contrast to the classical algorithm, Fig. 8b shows the results of using the DTW for dimensionality reduction. The distance matrices from (2) and DTW are fed to the MDS to achieve dimensionality reduction, respectively. In Fig. 8, each sample represents a chaotic time series and different colors represent different values of parameter ρ in (4). As can be seen from the figure, the difference between the results of the two methods is very obvious. Based on the proposed method, the data after dimension reduction are evenly distributed according to the change of parameter ρ. In common cases, some machine learning and data mining methods, such as classification and clustering, are used to evaluate the performance of dimension reduction of the original time series. For the Fig. 8a, the result is of great significance to machine learning and data mining. For example, we can use a simple linear classifier to get a good classification performance for this result. In contrast to the DTW-based results in Fig. 8b, the data are completely mixed together which cannot be distinguished according to the different values of parameter ρ. These results indicate that although DTW method has been widely used in measuring

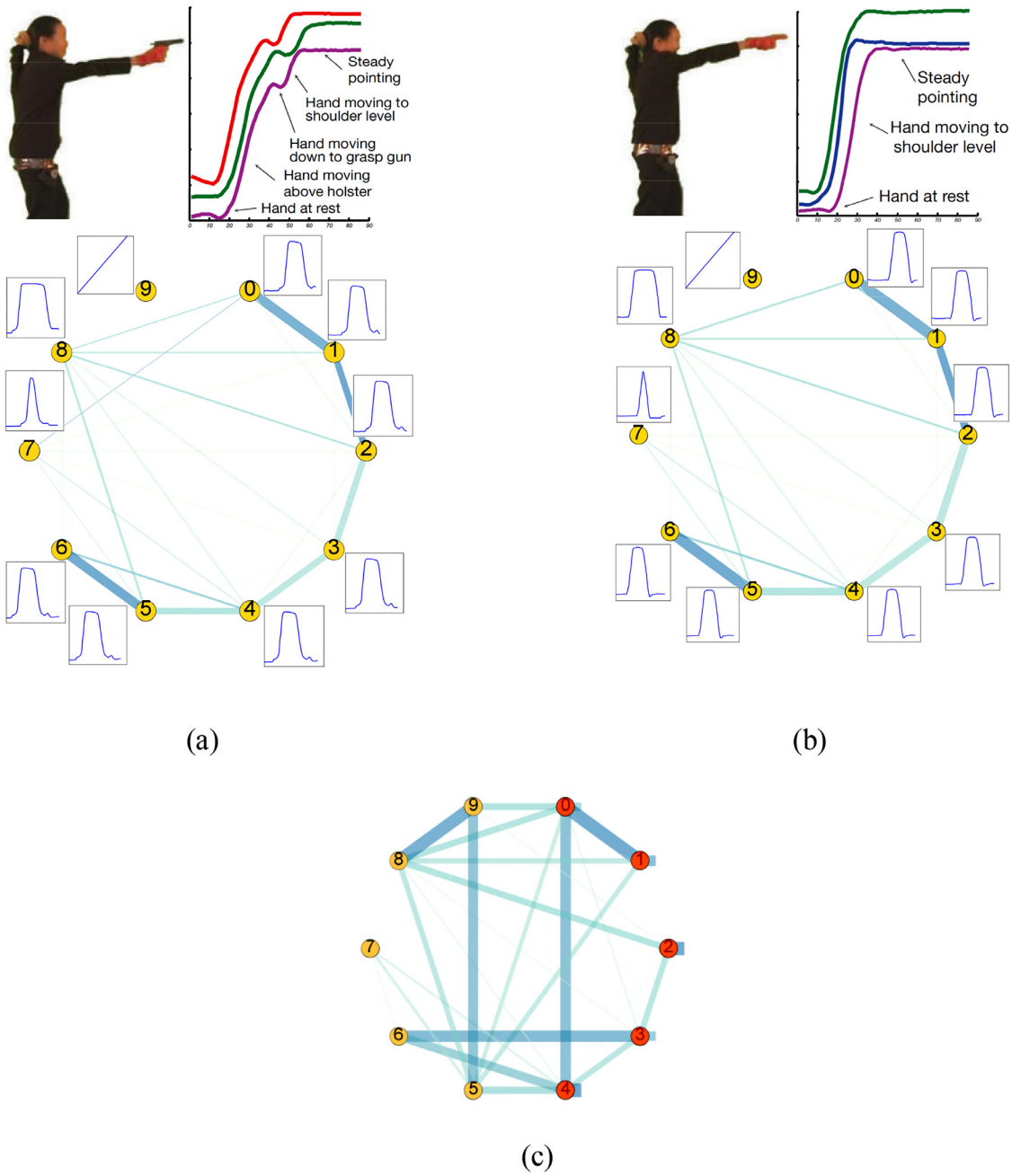


Fig. 10. Structural and functional networks of Gun-Point dataset. (a) Structural network of Gun class; (b) Structural network of Point class; (c) functional network. (We have got a permission from E.Keogh to use stills and annotation).

time series similarity, it cannot catch the similarity of the chaotic time series. The reason for the above results is that an extracted covariance matrix is a natural way to fuse the local features and their interactions. Thus the extracted features include not only the local features themselves but also their interactions. Furthermore, the global characteristics can be obtained by synthesis of the local features. In contrast, DTW only focuses on measuring local similarity. Due to the high complexity of the chaotic time series, it is not enough to consider only local similarity.

3.2. Classification accuracy of UCR time series

Four classes of previously competing classifiers are used for comparing the classification accuracy with LFI: best warping window DTW, no warping window DTW [38], and two state-of-the-art TSBF-based classifiers based on random and uniform subsequences [15]. Classification results for DTW-based classifiers were obtained directly from the project website [38]. For TSBF-based classifiers, the results were obtained by running the code provided by the author [48]. For LFI, 1-NN and SVM are trained in tangent space of the manifold. Based on 5-fold cross-validation on the training data, the parameter C in linear SVM is determined from the set {1, 10, 100, 1000}.

The classification accuracies are compared in Table 1 in which the best accuracy is denoted in bold. For the limited space, the table shows the results of 43 data sets which are selected from 85 data sets every one data set in alphabetical order. For each specific method, the best result of the two classifiers is used as the final result. For the DTW-based method, for example, we select the best accuracy between the best warping window DTW and no warping window DTW. The results in Table 1 show that LFI performs better than DTW-based and TSBF-based classifiers in 21 data sets. By

comparison, the numbers of best results of DTW-based and TSBF-based are 8 and 17, respectively.

To better visualize and evaluate the classification accuracy for all 85 data sets, we use the scatter plot proposed by Ding et al. to compare the results [49]. Fig. 9 presents pairwise comparisons of classification accuracy, where x axis represents the DTW-based or the TSBF-based method, y axis represents LFI method and each dot denotes the error rate for a particular dataset. The line $y = x$ represents the location where both methods perform the same. A dot above the line indicates that the method on the x axis is more accurate than that on the y axis of the corresponding data set. For 85 data sets, LFI beat DTW-based in 65 data sets and beat TSBF-based in 50 data sets. For some data sets, LFI exhibits large improvements in classification rate.

3.3. Analysis of classification by structural and functional network

To provide insight into the classifier, structural and functional networks of the Gun-Point dataset are shown in Fig. 10 as an example. The purpose of this data set is to identify whether a person draws a gun or not [50]. Thus it has two classes (Gun class and Point class) and corresponding structural networks of a sample are shown in Fig. 10a and Fig. 10b, respectively. Top panels show actor's stills and time series annotations for the two classes. Here embedding dimension $m = 7$ and time delay $\tau = 4$ so the number of nodes is $m + 3 = 10$. We can see that the topology structure of the two networks is similar except for the small differences in strength of the links. To avoid having too dense a graph, sparse inverse covariance is used for reducing the weak links [51]. As above mentioned, the nodes represent the subsequences which are drawn next to the nodes and the numbers inside the small circles denote the temporal order of subsequences; the links in structural network indicate relationships between nodes and the width represent the strength. It can be seen that the neighboring pairs have a strong physical relationship. This phenomenon is easy to understand as the neighboring subsequences share more information in common cases.

To illustrate the effect of the interaction between subsequences for classification, the functional network is derived by linear SVM and shown in Fig. 10c. We represent only the 30% links with the highest values to display clearly. It can be seen that its structure is quite different from the ones in Fig. 10a and b. The nodes and links here indicate the importance of nodes themselves and the interaction between them for classification. First, the red nodes {0,1,2,3,4} have more information than others and the significance is denoted by the width of an attached short bar. This phenomenon is easily explicable because there is a small peak just before the large peak in the middle for a Gun-Draw, namely, the difference mainly exists in the first half segment of a time series. Second, the interaction between pairs, such as {0,1}, {0,4}, {3,6}, {5,9} and {8,9} are key features for classification. It should be noted that covariation between the 9th node (temporal variable) and other nodes are also important factors for classification. In addition, some "hidden" interactions, such as {3,6} and {4,6}, are caught as distinguishing features. Though {3,6} has a weak physical connection, its interaction plays an important role to realize the classification.

4. Future work and conclusion

The key point of the proposed algorithm is to project the data from manifold to tangent space. As described in (7), projecting all points into a tangent space is an approximate method since Riemann manifolds are only homomorphic to Euclidean Spaces in local scale. Therefore, one way to improve the performance of the algorithm is to divide the data into different subsets to ensure that the manifolds are homomorphic to tangent spaces in local scale, then the subsets are projected into different tangent Spaces, and finally the vectors in these tangent Spaces are classified. Now, the key problem here is how to compare vectors in different tangent Spaces since each tangent space has its own coordinate system. In other words, all vectors must be in a common frame of reference, that is, they must be "transported" into a common tangent space. So our future work is to use affine connection and parallel translation to transform the tangent vectors in different tangent spaces into a common tangent space for comparison and classification. We believe that this can improve the performance of the algorithm to a certain extent and is a more general framework, especially when the distribution of data is diverse.

Based on the frame work of information geometry, a novel method is proposed for classifying the univariate time series. The covariance matrix is used as the representation of a univariate time series, which synthesizes the local and global characteristics of the time series. The structural and functional networks are also derived for interpreting the classifier. The structural network is used as the feature, and the functional network is used for explaining the importance of the interaction between the subsequences. We have demonstrated that the proposed approach achieves state-of-the-art performance and gives better results than competitive methods on a synthetic data and a set of benchmark data sets.

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